

## Professional Summary

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Experienced **Unity Developer** with over 7 years of hands-on experience delivering commercial games, VR simulations, and interactive software. Proven track record in shipping mobile games, building real-time EEG-integrated Unity visual systems, and optimising performance across VR and 3D interactive environments. Adept at combining technical execution with strategic thinking, managing both internal teams and client expectations. Comfortable working on end-to-end feature delivery, multiplayer networked environments, and immersive experiences.

## Core Skills

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- **Languages**  
C#, Rust, Dart, JavaScript, SQL, HTML, CSS
- **Game Engines & Platforms**  
Unity Engine, Unity VR, Netcode for GameObjects, FishNet  
Platforms: Mobile (iOS/Android), VR (Quest/PCVR), Web, Windows, Linux, Mac
- **Frameworks**  
ASP.NET, Entity Framework, React Native, Flutter, VueJS
- **CMS & Web**  
Umbraco, Kentico, Sitecore, MVC, Webforms
- **DevOps & Tools**  
Git, Azure DevOps, Jira, Docker, Unity Cloud Build, Octopus Deploy, TeamCity, XCode, Firebase, Trello

## Work Experience

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### Senior Unity Developer | Hive Mind Innovations | July 2024 – Present

- Designed and prototyped a real-time EEG visualisation system using Unity (Shader Graph, Particle System) and BrainFlow SDK in Python. Delivered an Innovate UK-funded MVP demo video combining EEG and beat detection data.
- Researched and implemented a custom beat detection algorithm in Rust for real-time audio analysis (frequency energy extraction, UDP integration with Unity). Enabled visuals to synchronise seamlessly with live music.
- Built a prototype mobile game in Unity integrating EEG controls via BrainFlow SDK and OpenBCI feeds. Used focus data to drive inputs, creating a tech demo for clients and laying foundations for a free-to-play spin-off.

### Lead Unity Developer | Spark Digital Studios | February 2022 – Present

- Developed and launched 8 casino games in Unity (Unity Cloud Build/Azure, Git, IronSource, Firebase, Nakama, Facebook SDK), reducing time-to-market by 15% while sustaining \$8M annual revenue and 99.8% crash-free sessions.
- Added live leaderboards and Facebook friend integration, boosting player engagement and retention.
- Designed and implemented a dynamic IAP store in Unity (C# with Firebase Remote Config & Analytics) to segment users and personalise purchase options, increasing ARPU by 20%.
- Built a seasonal asset-switching system (Unity Addressables, Firebase Remote Config, custom build tools) enabling design-driven updates, improving release speed, App Store visibility, and retention (+10%).
- Designed and delivered cross-promotion features (native iOS/Android APIs, Firebase Remote Config, custom build tools), driving a 20% increase in app installs during promotional campaigns.
- Designed a reusable high-speed RTP (Return to Player) simulation system, cutting test cycles from one week to <10 minutes while ensuring compliance accuracy across titles.
- Built VR prototypes including a space-station climbing simulator (using NASA assets) and a multiplayer-enabled VR office (VR Toolkit, UltimateXR, Netcode). Delivered demos to pitch VR capabilities to clients.
- Managed a cross-functional team through sprint planning, task assignment, and client-facing meetings.
- Line managed three developers, setting goals, running training, conducting code reviews, and mentoring career progression to improve delivery quality and team capability.

**Web Developer | Mosquito Digital | October 2019 – February 2022**

**Web Developer | Absurd | March 2019 – October 2019**

**Junior Web Developer | Ignite Insurance | October 2017 – March 2019**

## **Independent Unity Projects**

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- **Unity Dialogue System:** Built a node-based dialogue editor using Unity Graph View, enabling non-developers to create branching conversations.
- **Tower Defence RPG:** Developed a small-scale RPG with tower defence mechanics, economy systems, and story integration using the custom dialogue tool.
- **3D Modelling in Blender:** Re-engaged with Blender, modelling props like swords, floating islands, and stylised assets for use in personal game projects.